Aarhus university

Computer-Science

Numerical Linear Algebra

Handin 6

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Math done by a human

Consider the space of differentiable functions $f : [0, \pi] \to \mathbb{R}$ with the L^2 inner product

$$\langle f, g \rangle = \int_0^\pi f(x)g(x)\, dx.$$

Showing Orthogonality

For the given collection of functions

$$\sin(x), \sin(2x), \ldots, \sin(kx), \quad \text{For every integer } k > 0$$